

Bluestock Mutual Fund Analytics Capstone Project

1. Executive Summary

The Mutual Fund Analytics Capstone Project aims to analyze the Indian mutual fund industry using data-driven techniques. The project involved ETL processes, data cleaning, exploratory data analysis, fund performance evaluation, dashboard development, and advanced risk analytics. Interactive dashboards and analytical models were developed to provide meaningful insights into investor behavior, fund performance, SIP trends, and risk assessment.

2. Problem Statement

Investors often face difficulty in selecting suitable mutual funds due to the large number of schemes and the complexity of performance indicators. The objective of this project is to transform raw mutual fund data into actionable insights that can assist investors and financial analysts in decision-making.

3. Project Objectives

- Perform ETL operations on mutual fund datasets.
- Clean and preprocess the data.
- Conduct exploratory data analysis.
- Evaluate fund performance using financial metrics.
- Build an interactive Power BI dashboard.
- Implement advanced analytics and risk metrics.
- Develop a simple fund recommendation system.
- Present findings through reports and visualizations.

4. Data Sources

The project utilized multiple datasets including:

- Fund Master Data
- NAV History
- AUM Data
- SIP Inflows

- Category Inflows
- Industry Folio Count
- Investor Transactions
- Scheme Performance Data

These datasets were obtained as part of the Bluestock Mutual Fund Analytics Capstone Project.

5. ETL Design and Data Cleaning

The ETL process involved extracting raw datasets, transforming them through cleaning and validation, and loading them into structured formats for analysis.

Key activities:

- Handling missing values.
- Standardizing column names.
- Removing duplicates.
- Converting data types.
- Validating AMFI codes.
- Creating processed datasets.

6. Exploratory Data Analysis (EDA)

EDA was conducted to understand patterns and trends in the mutual fund industry.

Key analyses included:

- Distribution of AUM across fund houses.
- Category-wise inflow analysis.
- Monthly SIP trends.
- Investor demographics.
- Industry folio growth trends.

7. Fund Performance Analytics

Fund performance was evaluated using several financial metrics including:

- Alpha
- Beta
- Sharpe Ratio
- Sortino Ratio
- Standard Deviation
- Maximum Drawdown
- Benchmark Comparison

These metrics helped identify high-performing schemes across different categories.

8. Dashboard Development

An interactive Power BI dashboard was developed to visualize industry trends and investment insights.

Dashboard Pages:

- Industry Overview
- Fund Performance
- Investor Analytics
- SIP Market Trends

The dashboard enables users to explore key metrics and trends effectively.

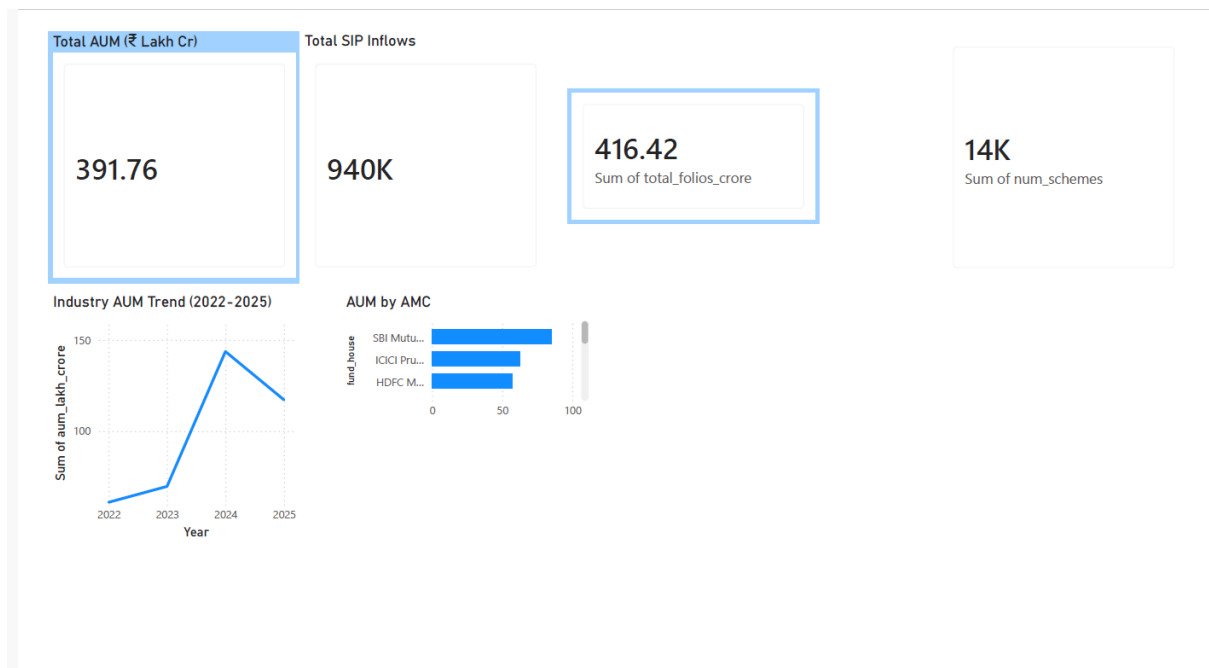


Figure 1: Industry Overview Dashboard

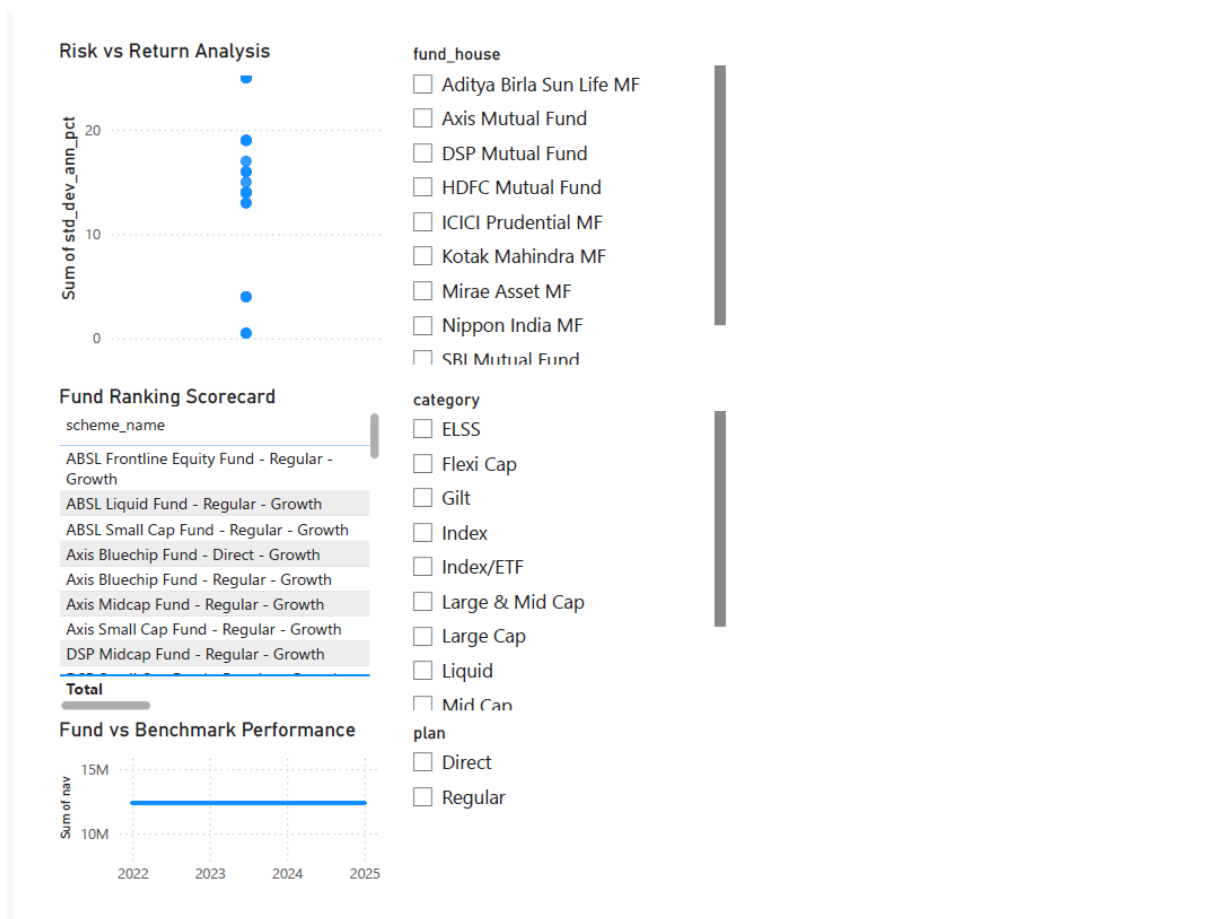


Figure 2: Fund Performance Dashboard

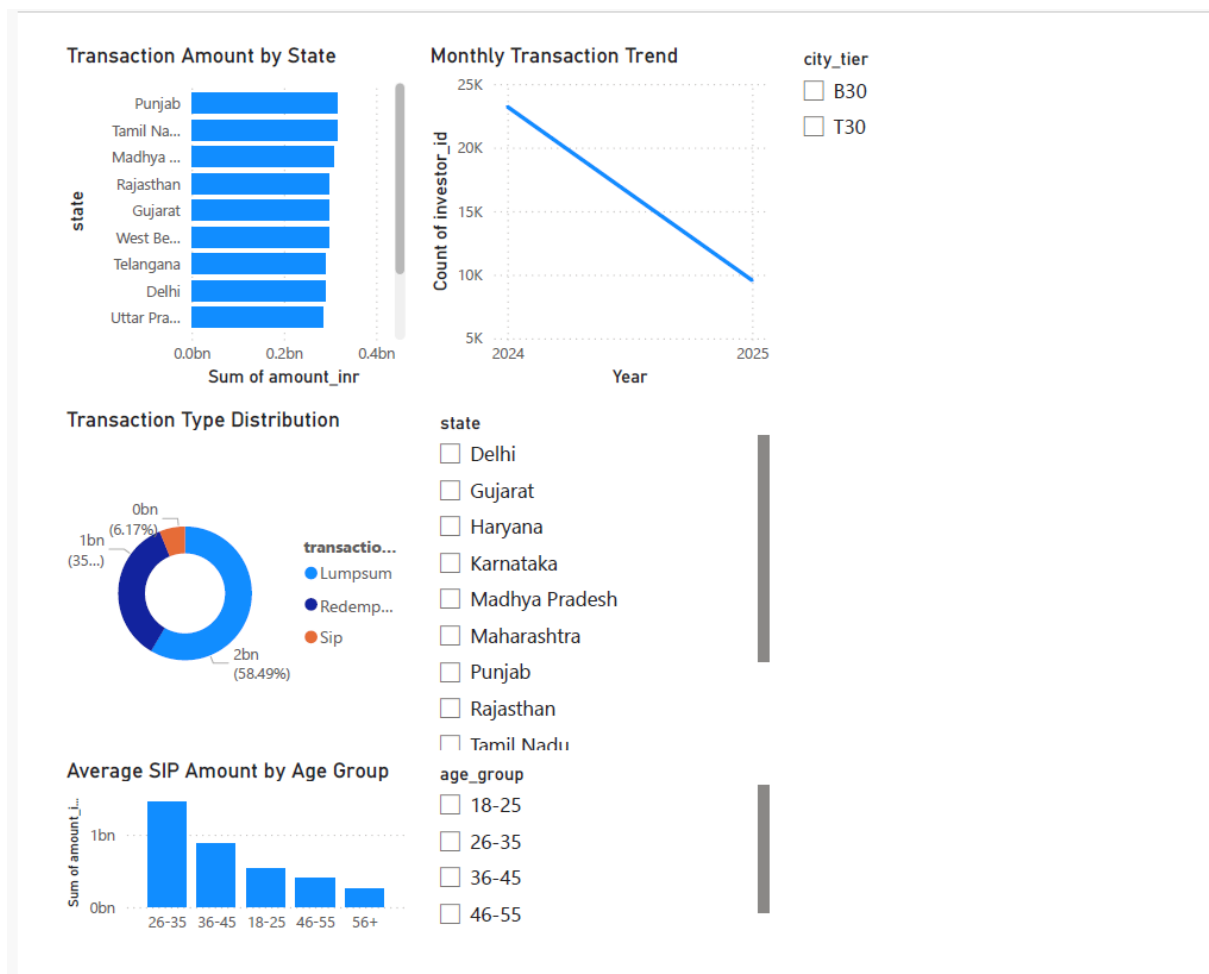


Figure 3: Investor Analytics Dashboard

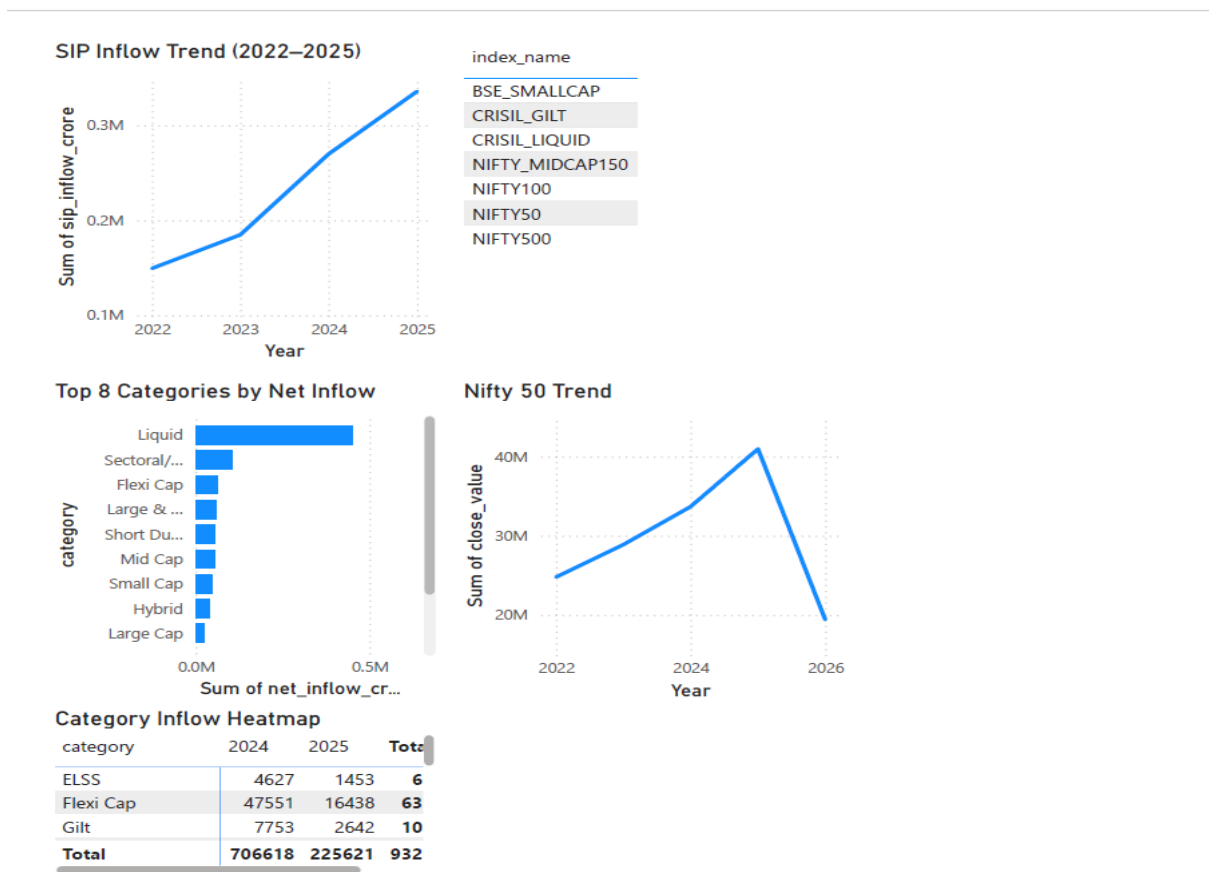


Figure 4: SIP Market Trends Dashboard

9. Advanced Analytics and Risk Metrics

Advanced analytical techniques were implemented to assess investment risk and investor behavior.

These included:

- Historical VaR (95%)
- Conditional VaR (CVaR)
- Rolling 90-Day Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- Simple Fund Recommendation System

10. Key Findings and Insights

- Certain mid-cap funds exhibited the highest Sharpe Ratios.

- SIP inflows demonstrated steady growth over time.
- Some investor cohorts contributed significantly higher investments.
- High-risk funds generated superior returns but carried greater downside risk.
- SIP continuity analysis identified investors potentially at risk of discontinuation.

11. Limitations

- Limited historical coverage of datasets.
- Recommendation system based on simplified logic.
- Absence of real-time market data.
- Sector allocation data limitations restricted deeper concentration analysis.

12. Recommendations

- Investors should diversify across risk categories.
- Regular monitoring of SIP continuity is recommended.
- Risk-adjusted metrics should be prioritized over absolute returns.
- Future models can incorporate machine learning techniques.

13. Conclusion

The project successfully transformed raw mutual fund data into meaningful business insights. Through analytics, dashboards, and risk evaluation, the project demonstrated how data science techniques can support investment decision-making in the mutual fund industry.

14. Future Scope

- Real-time data integration.
- Machine learning-based recommendation systems.
- Deployment of dashboards to cloud platforms.
- Personalized investor advisory systems.
- Predictive modeling for fund performance forecasting.